

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			Ammeter in series (1) Voltmeter shown in parallel with the thermistor (1) N.B. correct symbol shown in a circle	2			2		2
	(b)			6 points plotted correctly to within <1 small square tolerance (1) 5 points or less plotted correctly to within <1 small square tolerance (0) Curve of best fit between 0-125 °C <1 small square tolerance from plotted points (1)		2		2	2	2
	(c)	(i)		Total resistance = $40 + 40 = 80 \text{ } [\Omega]$ (1) so $I = \frac{12}{80} = 0.15 \text{ [A]}$ (1) Therefore $V = 0.15 \times 40 = 6 \text{ [V]}$ (1) Alternative: Voltage shared equally (1) so $V = \frac{12}{2} \text{ (1)}$ $= 6 \text{ [V]}$ (1)		3		3	3	3
		(ii)		As temperature increases, {the resistance of the thermistor decreases/ the resistance shown by the {graph / table} decreases (1) Total [series] resistance decreases, [so current increases] (1) [so agree]			2	2		2